

Endorsement Policies in Hyperledger Fabric

Chaincodes in Hyperledger Fabric must be endorsed by peers, else they remain invalid. Endorsement requires policies that have been set up at the time of instantiation or upgradation. This article is designed for open source technology enthusiasts who have prior experience with Hyperledger Fabric.



Every chaincode has an endorsement policy, which specifies the set of peers on a channel that must execute chaincode and endorse the execution results in order for the transaction to be considered valid. These endorsement policies define the organisation's peers who must endorse, i.e., approve of the execution of a proposal. As part of the transaction validation step performed by the peers, each validating peer checks to make sure that the transaction contains the appropriate number of endorsements and that they are from the expected sources.

Endorsement in Hyperledger Fabric

By default, endorsement policies are specified for a channel's chaincode at the time of instantiation or upgradation, i.e., one endorsement policy covers all of the states associated with a chaincode. However, there are cases where it may be necessary for a particular state (a

particular key-value pair, in other words) to have a different endorsement policy. This state based endorsement allows the default chaincode-level endorsement policies to be overridden by a different policy for the specified keys.

Chaincode-level endorsement

Chaincode-level endorsement policies can be specified at the time of instantiation using either the SDK or, in the peer CLI, the `-P` switch followed by the policy. For example:

```
peer chaincode instantiate -C <channelid> -n mycc -P  
"AND('Org1.member', 'Org2.member')"
```

This command deploys *chaincode mycc* ("my chaincode") with the policy `AND ('Org1.member', 'Org2.member')` which would require that a member of both Org1 and Org2 sign the transaction. If the identity classification is enabled,